

DU Mathsoc Problem Solving: Set 2

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~Advanced Paper~

For any assistance or inquiries about the problem sets, please email me at: quizmaster.maths@csc.tcd.ie.

Problem 1. How many n -digit numbers whose digits are in the set $\{2, 3, 7, 9\}$ are divisible by 3?

Problem 2. A 6×6 grid is tiled by dominoes. Prove that there exists some line which cuts the board into two nonempty parts without cutting through any domino.

Problem 3. Choose four points at random on the surface of a sphere. Find the probability that the centre of the sphere is contained within the convex hull of the four points.

Problem 4. Let (X, d) be a compact metric space and $(U_i)_{i \in I}$ be an open cover of X . Show that there exists $\varepsilon > 0$ such that, for every $x \in X$, the open ball $B(x, \varepsilon)$ is contained in some member of the cover.

Problem 5. Let G be a group. Prove that for every $a \in G$, the map from G to G given by $x \mapsto axa^{-1}$ is an isomorphism.